

QINGHAI JIANG

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🐙 <https://github.com/andy-qhjiang>

Education

- 2024-present 📖 **PhD in Applied Mathematics, The Chinese University of Hong Kong**
Advisor: Gary Pui Tung CHOI
- 2021-2024 📖 **Master in Science, Fudan University**
Thesis title: *Ramanujan-Mordell Type Formulas Associated To Certain Theta Function.*
Advisor: Rufe Ren
- 2017 – 2021 📖 **Bachelor in Mathematics and Applied Mathematics , Sun Yat-Sen University.**
Thesis title: *Proof of some new theta function relations.*
Advisor: Dongxi Ye

Honors and Awards

- 2023 📖 **Merit Award**, Fudan University.
Maxwealth Fund Scholarship, Maxwealth Fund Management Co.
- 2022 📖 **Merit Award**, Fudan University.
China Pacific Insurance Scholarship, China Pacific Insurance Co.

Research Publications

Published

- 1 Q. Jiang and G. P. Choi, "Pykirigami: An interactive python simulator for kirigami structures," *Computer Physics Communications*, vol. 328, p. 110 349, 2026, ISSN: 0010-4655. 🐙 DOI: <https://doi.org/10.1016/j.cpc.2026.110349>.
- 2 Y. Ding, D. Gu, T. Huang, Q. Jiang, and D. Ye, "On conjectures of z.-h. sun," *Results in Mathematics*, vol. 77, no. 3, p. 105, Apr. 2022, ISSN: 1420-9012. 🐙 DOI: 10.1007/s00025-022-01640-5.

Preprints

- 1 Q. Jiang and G. P. T. Choi, *A unified geometric design framework for kirigami structures*, 2026. 🐙 URL: <https://arxiv.org/abs/2608.30032>.

Manuscript in preparation

Teaching Assistant Experiences

- 2026.1–2026.5 📖 Mathematics Laboratory (MATH2221).
- 2025.9–2025.12 📖 Mathematics of Multivariate Data (MMAT5396).
- 2025.1–2025.5 📖 Mathematics Laboratory (MATH2221).
- 2024.9–2024.12 📖 Social and Economic Networks (MMAT5370).

Research Project

2024 – present

- **Computational Kirigami Design via Nonlinear Optimization.** Developed a MATLAB-based framework for inverse design about kirigami patterns — 2D cut sheets that deploy into prescribed 2D curve or 3D surfaces through rigid quad rotation. The optimization problem (implemented via IPOPT) enforces edge-length isometry as a hard constraint while minimizing deviation from the target surface, with overlap-prevention inequalities ensuring physical realizability. The pipeline supports diverse target geometries (spheres, saddles, tori, Möbius strips) and outputs patterns directly compatible with a custom physics simulator (PyKirigami) for deployment validation.
- **Kirigami Deployment Simulation.** Developed PyKirigami, an open-source 3D rigid-body simulator for kirigami mechanism deployment, built on PyBullet's maximal-coordinate formulation with an impulse-based constraint solver. Validated kinematic accuracy against analytical solutions, achieving sub-percent vertex error throughout deployment. Conducted systematic sensitivity analysis of actuation parameters and benchmarked computational performance against Pymunk.

2021 – 2022

- **Some equations about the coefficients of newforms.** Use elliptic curve and newforms to prove some equations seen in my Github.

Software

Metamaterials

- PyKirigami: <https://github.com/andy-qhjiang/PyKirigami>